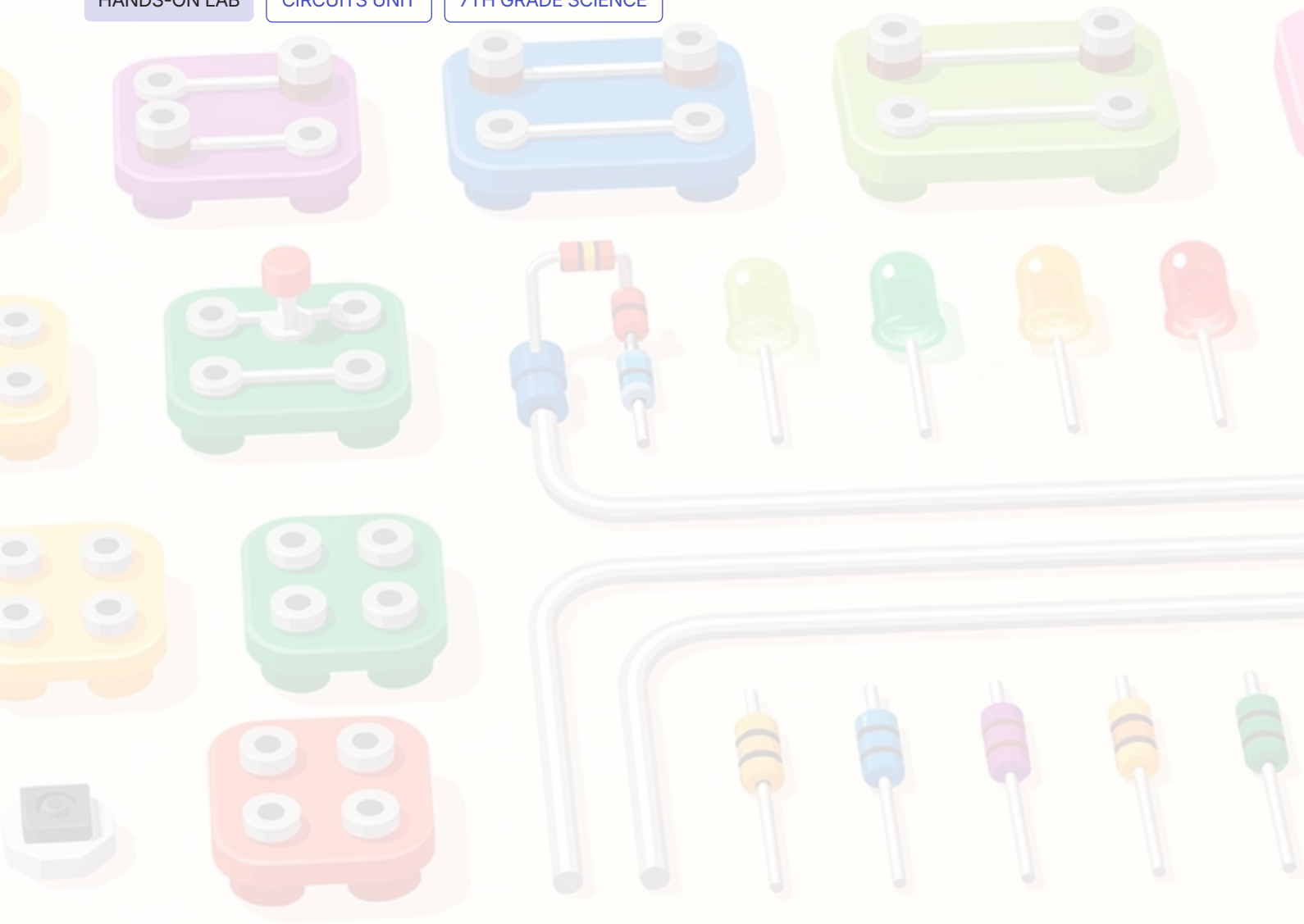
⚡ Snap Circuits Live! From Virtual to Real

You just built circuits on a screen — now we're going to **make electricity actually DO something** with real components you can hold, snap together, and test. Get ready to go from pixel-perfect to hands-on awesome. This lesson takes everything you figured out in PhET and levels it WAY up.

HANDS-ON LAB | CIRCUITS UNIT | 7TH GRADE SCIENCE



🔄 Transition: From PhET to the Real World

> 🗣️ **Teacher Script — Say This Out Loud:**
> "Okay, so in PhET we were basically playing circuit architect on a computer — dragging wires, dropping bulbs, watching electrons flow. That was awesome because we could experiment without breaking anything or shocking ourselves. But here's the thing: real circuits? Real electrons? Real components? It's a completely different experience. Like the difference between watching a cooking video and actually making the food. Today, we snap things together for real. Same rules apply — voltage, current, series, parallel — but NOW you're going to feel it, see it, hear it, and honestly... probably be surprised a few times."

What You Did in PhET

- Dragged virtual wires and components
- Tested series and parallel circuits safely
- Measured voltage and current with virtual meters
- Saw electron flow animations
- Could "undo" any mistake instantly

What You'll Do with Snap Circuits

- Snap real components onto a grid board
- Use actual electrons moving through wires
- Observe real light, sound, and motion
- Troubleshoot when things DON'T work
- Experience the real engineering design process

The big idea? **Everything you learned on PhET is still true.** Electricity still needs a complete loop. Voltage still pushes current. Resistance still limits flow. We're just trading the mouse for our fingers and the screen for a snap board. Let's go! ⚡

📋 Today's Roadmap — What We're Building Toward

This isn't just "plug stuff in and hope it lights up." We're going on a structured journey that builds your understanding layer by layer. By the end, you'll have built five different circuit projects, understand how to read a snap circuit diagram, and be ready to program your first microcontroller. Here's the full game plan:

01

Safety Check + Component ID

Learn the rules, name every part, and set up your workspace like a real electronics engineer.

02

Build Project 1 — Simple LED Circuit

Your first real circuit. No tricks, just power + LED + switch. Prove electricity needs a loop.

03

Series vs. Parallel Challenge

Build both types, compare them side-by-side, and answer the big question: which one do real houses use and why?

04

Snap Circuits + Sound — The Fan and Alarm

Add a motor and buzzer. Discover how electricity creates motion and sound.

05

Easter Egg Light Challenge + Capstone Reflection

Creative challenge using the Easter Egg Light set. Design your own light sequence and document your work.

> 📷 **Capstone Note:** We'll flag specific moments throughout this lesson where you should take a photo for your capstone presentation. Watch for the 📷 symbol!

🛡️ Safety First — The Four Rules of Snap Circuits

Before we touch a single snap, we do safety. I know, I know — "it's just a toy kit." Nope. It's a real electronics kit that uses real electricity. The batteries are low voltage, so you're not going to get hurt badly, but you CAN damage components, cause short circuits, or get a surprising little zap if you're not paying attention. Real engineers follow safety protocols every single time, no exceptions.

Rule 1: No Loose Metal

Never let bare metal wires touch each other outside of a snap connection. That's a short circuit — instant battery drain and possible heat.

Rule 2: Battery In LAST

Always build your complete circuit BEFORE inserting batteries. Check your layout, THEN power it up. This prevents accidental shorts.

Rule 3: If It Gets Hot — STOP

If any component feels warm or you smell anything unusual, disconnect the battery immediately and tell me. Heat = something's wrong.

Rule 4: Snap, Don't Force

Components snap onto the grid gently. If you're pushing hard, something is misaligned. Forcing connections can break snaps permanently.

🗣️ **Teacher Script:** "Before we do ANYTHING, tell me — what do you think would happen if two wires touched each other and bypassed the LED? Would the LED get brighter? Dimmer? Neither? Let's think about this..." **[Wait for student answer]**

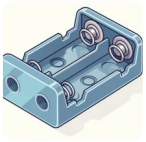
💡 **Possible Student Answers:**

- "The LED would get brighter" (common misconception — more power going through)
- "Nothing would happen" (unsure about short circuits)
- "The LED would go out / get dimmer" (partially correct reasoning)
- "The battery would drain faster / get hot" (strong answer)

Teacher Follow-up: Guide students toward: bypassing the LED means current takes the path of least resistance — the LED gets no current and goes out, while the battery drains rapidly and may heat up. This is exactly what a short circuit is.

🔍 Component Identification — Name That Part!

Real engineers don't say "the thing that lights up" — they use precise vocabulary. Let's do a rapid-fire component ID round. Pull out all the components from your sets and lay them on the table. We're going to identify every single one before we build anything. This is your toolkit — know it cold.



Battery Holder (B1)

The power source. Holds AA batteries. This is your "voltage source" — same as the battery symbol you used in PhET. Connects to the positive rail on the snap board.

LED (D1)

Light Emitting Diode. Converts electrical energy to light. Has polarity — must be connected in the correct direction or it won't light up. The longer leg is positive (+).

Switch (S1)

Opens and closes the circuit. Flip it one way = complete circuit = current flows. Flip the other way = open circuit = current stops. You controlled this virtually in PhET!

Motor (M1)

Converts electrical energy into mechanical energy (spinning). Found in your Snaptricity and Skill Builder sets. Wait until you see how fast this thing spins!

✏️ **Note-Writing Prompt:** In your science notebook, create a two-column table. Left column: "Component Name + Symbol." Right column: "What it does in a circuit (in your own words)." Fill in at least 8 components from your sets. Use the component guide from the kit as reference — but write the definitions in YOUR words, not copied from the guide.

🎨 Draw It! — Your Snap Circuits Parts Diagram

Now we're doing something that real electrical engineers do every single day: we're making a component reference sheet. This isn't busywork — when we get into complex circuits, you'll thank yourself for having this. Plus, drawing something forces your brain to actually look at it carefully, which helps you remember it WAY better than just reading a list.

✏️ **Drawing Prompt — Do This Now:** Get out a piece of white paper, a pencil, and colored pencils. Title "Component Reference Sheet". Draw and label the following components. For each one, draw: (1) what it actually looks like as a snap component, and (2) the electrical symbol we used in PhET. Components to draw: Battery, LED, Switch (open AND closed), Resistor, Buzzer, Motor.

Now list all of these items in your digital notebook. Leave space next to each symbol to add notes as we use them today. These pages will become your personal "cheat sheets" for the rest of the circuits unit.

Snap Component Look

Draw the actual physical piece — the colored plastic rectangle with snap connectors on it. Note the color of each component (batteries are usually red/black, LEDs are clear or colored, switches are often white or yellow).

- Show the snap connectors on top and bottom
- Label the positive and negative sides where visible
- Write the component code (B1, D1, S1, etc.)

Circuit Symbol Version

Next to each physical drawing, write the standard electrical symbol from PhET. This is the bridge between the virtual and physical world — the same component, two different representations.

- Use a ruler for straight wire lines
- Mark the + and – ends on your battery symbol
- For the light component, draw BOTH symbols side by side: the PhET lightbulb symbol (a circle with a x inside) AND the real-world LED symbol (a triangle pointing to a line, with two small arrows showing light coming out). Label them: 'Lightbulb symbol' and 'LED symbol'. Same job, different look!

📷 **Capstone Photo Moment!** Once you finish your component reference page, take a photo of it. This is a great "before" shot for your capstone — it shows what you knew at the START of the lesson. Label this photo: "Component Reference Sheet — Beginning of Snap Circuits Lesson."

⚡ Project 1 — Your First Real Circuit: Simple LED Loop

Alright — moment of truth. We're building your very first real circuit. This is the "Hello World" of electronics — simple, satisfying, and it tells you immediately whether you understand the most fundamental rule in all of electricity: **current needs a complete, unbroken loop to flow.** No loop = no light. Complete loop = light. That's it. That's the whole game right now.

> 👀 **Teacher Script:** "Before you snap a single piece, I want you to predict: what's the minimum number of components you need to light up one LED? Don't look at the kit guide. Just think about what electricity needs. Talk me through it." **[Expected answer: battery + LED + wires to complete the loop.]**

Use components from your **Snap Circuits Junior 100** set for this project. You need: Battery Holder (B1), one LED (D1), one Slide Switch (S1), and two short wire jumpers. Follow the diagram above, then insert batteries only after your circuit layout is complete and checked.

1

Place the Battery Holder

Set the battery holder (B1) at the top of your snap grid. The red wire side is positive (+), black is negative (−). Orient it so positive is on the left rail. Don't put batteries in yet!

2

Add the Switch

Snap the slide switch (S1) into the circuit path coming from the positive terminal. Make sure the switch is in the OFF position before continuing. This gives you control over when current flows.

3

Connect the LED

Snap the LED (D1) into the circuit. Remember: LEDs have polarity! The longer leg (anode) connects toward the positive side. If you get it backward, it simply won't light — no damage, just reverse it. When current flows through the LED in the correct direction, electrons inside the semiconductor material lose energy by dropping to a lower energy level — and that released energy becomes a photon, a tiny packet of light. That's literally what you're seeing when it glows.

4

Complete the Loop

Run a wire from the LED back to the negative terminal of the battery holder. Trace the entire path with your finger to confirm it's a complete loop. NOW insert batteries. Flip the switch. 💡

5

💡 How Does Light Actually Happen?

When current flows through the LED, electrons inside the semiconductor material drop to a lower energy level — and that released energy becomes a photon, a tiny packet of light. The color depends on how big the energy drop is, determined by the LED's material. Red LEDs drop less energy than blue ones. You're not just completing a circuit — you're watching electrons release energy as light, one photon at a time. We will discuss more about colors later.

🤔 Project 1 — Reflection Q&A

Your LED is lit. Awesome. But we're not done with Project 1 yet — now we think. Real scientists don't just do an experiment and move on. They analyze what happened, ask new questions, and push the understanding deeper. This is your moment to connect your PhET experience to what you're seeing in front of you right now.

❓ Q1: What happens when you flip the switch off?

Expected Answer: The LED turns off. **Follow-up to ask:** "Where is the circuit broken when the switch is off? Can you point to the exact gap?" The student should be able to identify that the switch creates a physical break in the conductive path, stopping current flow — just like the open switch in PhET.

❓ Q2: What if you flip the LED backwards?

Expected Answer: It won't light up. **Follow-up:** "Does it break the circuit entirely? Is current still trying to flow?" This introduces the concept of diode behavior — LEDs only allow current in one direction. This is different from a regular resistor! Ask: "What does the 'D' in LED actually stand for?"

❓ Q3: What do you think would happen if we added a second LED in the SAME path?

Expected Answer: Student will guess — accept any prediction. **This sets up the next project!** Tell them: "Hold onto that prediction. We're about to find out. And I'll bet what actually happens surprises you a little bit."

- ✏️ **Note-Writing Prompt:** Write 3–4 sentences in your notebook answering this: "What does electricity need to flow in a circuit? Use the words: voltage, current, complete loop, and switch. Also write your prediction for what will happen when we add a second LED. Finally, write a one-sentence definition of the word photon in your own words — what is a photon and how does it connect to the LED you just lit up?"

  📝 Example: *"Electricity needs voltage to push it and a complete loop to travel through. Current flows from the battery, through the switch, through the LED, and back to the battery. If the switch is open, the loop is broken and current stops. I predict that adding a second LED will make both LEDs dimmer because they have to share the voltage. A photon is a tiny packet of light energy that is released when electricity flows through the LED and excites the electrons inside it."*

- 📷 **Capstone Photo Moment!** Take a photo of your completed Project 1 circuit with the LED glowing. This is your proof of first real circuit! Label it: "Project 1 — Simple LED Loop, [your name], [date]."

🔧 The Troubleshooting Toolkit — When It Doesn't Work

Here's something no one tells you enough: **things not working is completely normal in engineering.** Professional engineers spend a huge percentage of their time debugging, troubleshooting, and fixing circuits. In fact, the ability to systematically troubleshoot is considered one of the most valuable skills an electronics engineer has. So when your circuit doesn't light up — and at some point today, it won't — here's your systematic approach to fixing it.

1 Check the Battery First

Are the batteries actually inserted? Are they fully charged? Are they correctly oriented (+ to + red wire)? Dead or backwards batteries are responsible for at least 30% of "broken" circuits in this class. It's always worth checking first, even if you're sure you did it right.

2 Trace the Complete Loop

Put your finger on the positive battery terminal and physically trace the entire circuit path, snapping connection by connection, all the way back to the negative terminal. If your finger ever "falls off" the circuit — that's where the break is. An open circuit is literally just one spot where the loop is broken.

3 Check Polarity on LEDs

Flip the LED around. This is the #1 most common mistake with Snap Circuits. LEDs are one-way components. If it's backward, it blocks current in that direction. Simply reversing it costs you zero time and fixes the problem instantly when this is the issue.

4 Verify All Snaps Are Fully Seated

Press each snap connection firmly and feel for the click. A connection that LOOKS good but isn't fully snapped creates a high-resistance junction that can dim components or block current entirely. Wiggle each piece slightly — it shouldn't rock around.

5 Simplify and Test Sections

If you can't find the problem, strip the circuit down to the simplest version (just battery + one LED, no switch, no resistor). If THAT works, add components back one at a time until it stops working — that's your culprit. This is called "divide and conquer" debugging and it works every time.

✏️ **Note-Writing Prompt:** Write the 5 troubleshooting steps in your notebook in your own words, like a quick-reference card.

If you think of other ideas, you can list them here as well!

🔗 Project 2 — Series vs. Parallel: The Big Showdown

This is the most important project of the day. If you understand series versus parallel after this, you understand something that literally every electrician, electrical engineer, and electronics hobbyist on the planet has to know cold. We did this on PhET, and now we're going to build BOTH types with real components and compare them directly — side by side. Get ready because the difference is going to be obvious and kind of shocking (pun intended).

🗣️ **Teacher Script:** "Quick — without looking at your notes — what's the key difference between a series circuit and a parallel circuit? Don't give me a textbook definition. Describe it to me like you're explaining it to a friend who never heard of either one." **[Listen for: series = one single path, parallel = multiple paths. Correct any misconceptions before building. This is a great diagnostic moment.]**

💡 **Possible Student Answers:**

- "Series is one path, parallel is two paths" (correct — this is the target answer)
- "Series is when things are connected end to end" (correct phrasing — affirm it)
- "Parallel is when things are side by side / branching" (correct — great visual description)
- "I think series means they all share the same wire?" (partially correct — guide toward the single-path concept)
- "Parallel means if one breaks the others still work" (correct consequence — but push them to explain WHY)

🔴 Build A: Series Circuit

Using your Snap Circuits Junior 100 set:

- Battery holder (B1)
- Switch (S1)
- LED #1 (D1) snapped in the main path
- LED #2 (D2 or second LED) snapped AFTER the first one, in the same path
- Wire completing the loop back to negative

Both LEDs are in ONE single path. Current must flow through LED #1 to get to LED #2. Build this, then observe: How bright are the LEDs compared to just one?

🔵 Build B: Parallel Circuit

Using your Snaptricity or Skill Builder set:

- Battery holder (B1)
- Switch (S1)
- LED #1 (D1) in one branch
- LED #2 in a SEPARATE branch alongside the first
- Both branches connecting back to the same negative terminal

Each LED has its OWN path back to the battery. Current splits and travels through both paths simultaneously. Build this and compare brightness to Build A!

📊 Series vs. Parallel — Data Collection & Analysis

Now that you've built both circuits, it's observation time. We're going to be systematic about this — just like a real scientist. We don't just say "it looked brighter." We describe, compare, measure where we can, and draw conclusions. Use the table below as a guide for your notebook entry.

1	2+	↓	=
Path in Series	**Paths in Parallel**	**Series Brightness**	**Parallel Brightness**
Only one route for current to travel through all components	Multiple routes — current splits and travels simultaneously	Each LED in series gets a fraction of the total voltage — dimmer!	Each LED in parallel gets the FULL battery voltage — same as one LED alone!

✏️ **Data Table Prompt:** Draw a two-column comparison table in your notebook with these rows: (1) Number of paths for current, (2) LED brightness compared to single LED, (3) What happens when you remove ONE LED, (4) Which type uses more total battery power, (5) Where is this used in real life? Fill in both columns (Series / Parallel) as we discuss.

📢 **Teacher Script — The "Remove One LED" Test:** "Okay here's the BIG demonstration. I want you to carefully remove one of the LEDs from your Series circuit. What happens to the other one? Now do the same with your Parallel circuit. Ready? GO." **[Series: both go out. Parallel: the other one stays lit!]**

"THERE IT IS! That right there is why houses are wired in parallel. If one lightbulb burns out in your kitchen, do all your other lights go out? No! Because every outlet and light fixture in your house is on its own parallel branch."

📷 **Capstone Photo Moment!** Take a photo showing BOTH circuits side by side — one series, one parallel — both lit up. This is a powerful visual comparison for your capstone. Label it: "Series Circuit (left) vs. Parallel Circuit (right) — brightness comparison."

🧠 Quick Quiz — Series & Parallel Check

No pressure, no grade — this is just a quick knowledge check to make sure your brain is processing things correctly before we build something more complex. Answer out loud or write them down, whichever you prefer. I'll tell you right away if you're on track.

1

Question 1

A string of old-fashioned Christmas lights is wired in series. When ONE bulb burns out, ALL the lights go out. Using what you just observed, explain exactly why this happens in terms of current flow.

2

Question 2

In a parallel circuit with two branches, if one branch has a higher resistance than the other, which branch will have MORE current flowing through it? Hint: think back to Ohm's Law from PhET — V = IR. What does that mean for current when resistance goes up?

3

Question 3

You want to wire up a bedroom with three lamps. Should you wire them in series or parallel, and why? Give at least TWO reasons for your choice based on what you've observed today.

4

Bonus Challenge

In your series circuit with two LEDs, each LED was dimmer than with just one LED. Where did the "missing" brightness go? (Hint: the energy didn't disappear — it's a conservation of energy question. What was the voltage doing differently?)

Teacher Notes for Answers: Q1: The broken bulb creates an open circuit — the single path is broken, so no current can reach any other bulb. Q2: Lower resistance branch gets more current (I = V/R, so as R decreases, I increases). Q3: Parallel — each lamp gets full voltage AND removing one doesn't affect the others. Bonus: The voltage is split across the two LEDs in series (Kirchhoff's Voltage Law — voltages in a series loop add up to the source voltage).

🌀 Project 3 — Adding Motion: The Motor Circuit

So far we've made light. Now we're going to make something **move**. A motor converts electrical energy into mechanical energy — it's the same principle behind every electric car, every fan, every hard drive, and every robot you've ever seen. When you add a motor to a circuit, you're bridging the worlds of electricity and mechanics in a really satisfying way. This one is genuinely fun to play with.

🗣️ **Teacher Script:** "So remember when we talked about photons — how the LED was releasing little packets of light energy? That's electromagnetic energy. Electrical energy going in, light energy coming out. Now we're going to do something totally different with electricity. Instead of making light, we're going to make something physically spin. Before we build it, I want you to think about energy transformations again. In the LED circuit, what was the energy transformation? [electrical → light + heat] Now what do you predict will happen when we replace the LED with a motor? [electrical → mechanical/kinetic energy] Good. Let's build it."

💡 **Possible Student Answers:**

For "What was the energy transformation in the LED circuit?":

- "Electrical energy → light" (correct, but incomplete — prompt: "What else? Touch the LED carefully...")
- "Electrical energy → light and heat" (excellent — full answer)
- "Battery energy → light" (acceptable — guide toward using "electrical energy" as the precise term)

For "What will happen when we replace the LED with a motor?":

- "It will spin / move" (correct — guide to: what TYPE of energy is spinning?)
- "Electrical energy → mechanical energy / kinetic energy" (excellent — affirm this)
- "It'll make the fan go" (correct observation — push for the energy transformation language)
- "Will it go fast?" (redirect: "Great question — let's find out. But first, what's the energy transformation?")

1

Components Needed

From Snaptricity set: Battery holder (B1), Switch (S1), Motor (M1), Fan blade attachment, connecting wires. Optional: add a resistor (R1) to slow the motor down.

2

Build the Circuit

Wire: Battery (+) → Switch → Motor → Battery (−). This is a simple series circuit just like Project 1, but replace the LED with the motor. Snap the fan blade onto the motor shaft.

3

Test and Explore

Insert batteries. Flip the switch. Observe the fan speed. Now try: (a) reverse the motor's connections — which direction does it spin? (b) add a resistor in series — what happens to speed?

✏️ Project 3 — Reflection & Notes

✏️ Note-Writing Prompt

Write a short paragraph (4–6 sentences) describing the energy transformations in the motor circuit. Use these sentence starters to build your paragraph:

1. In the motor circuit, chemical energy stored in the batteries is transformed into...
2. This electrical energy travels through the wires and reaches the motor, which converts it into...
3. You can tell this is happening because...
4. The circuit also releases a small amount of ___ energy, which you can notice by...
5. When I added a resistor to the circuit, the motor slowed down because...

💡 *Challenge: Can you write your paragraph without using the starters? Try it!*

📝 Example of a strong response:

1. In the motor circuit, chemical energy stored in the batteries is transformed into **electrical energy that flows through the wires.**
2. This electrical energy travels through the wires and reaches the motor, which converts it into **mechanical (kinetic) energy, making the fan blade spin.**
3. You can tell this is happening because **the fan blade starts spinning as soon as you flip the switch.**
4. The circuit also releases a small amount of **heat** energy, which you can notice by **carefully touching the motor after it runs for a while — it feels slightly warm.**
5. When I added a resistor to the circuit, the motor slowed down because **the resistor reduced the current, giving the motor less electrical energy to convert into motion.**

🔊 Project 4 — Making Noise: The Buzzer Alarm Circuit

Let's add a new component to our toolkit: the buzzer. A buzzer converts electrical energy into sound waves. It does this by rapidly vibrating a small metal disc (a piezoelectric element) that pushes air back and forth, creating pressure waves that your ears detect as sound. It's kind of amazing when you think about it — electrical oscillation becoming literal vibrations in the air. This project is going to be hands-down the loudest thing we do today, fair warning.

🔧 Build: Alarm Circuit

Using Skill Builder set components:

- Battery holder (B1)
- Slide switch (S1)
- Buzzer (SP1 — speaker/piezo)
- Connecting wires

Wire them in series just like Project 1. The buzzer will sound when the switch is closed. Now add the LED from Project 1 — buzzer AND light at the same time!

🧠 Extend It — The Intruder Alarm

Now here's a real challenge: can you wire the buzzer and LED in PARALLEL instead of series? What changes about the sound volume and light brightness compared to the series version? Try it and document the difference.

Predict first: If you add the LED in parallel with the buzzer, will the buzzer be louder, quieter, or the same? Why? After building it, evaluate your prediction — were you right?

🗣️ **Teacher Script — Energy Conversion Check:** "Hold on — before you flip that switch, I need you to do three things: (1) Tell me the energy conversion that's about to happen. (2) Predict how the sound will change if I put a resistor in series with the buzzer. (3) Tell me whether the buzzer cares about polarity like an LED does, or not — and how you'd test that." **[Let the student work through this reasoning. Then let them test it.]**

💡 **Possible Student Answers:**

(1) "Tell me the energy conversion that's about to happen":

- "Electrical energy → sound" (correct)
- "Battery → buzzing / noise" (acceptable — guide toward formal energy language)
- "Electrical energy → sound and heat" (excellent — very thorough)

(2) "How will the sound change if I put a resistor in series with the buzzer?":

- "It'll get quieter" (correct — less current = less vibration amplitude)
- "It won't change" (misconception — guide: resistor reduces current, which affects the buzzer's power)
- "It might stop working" (possible if resistance is very high — good critical thinking)

(3) "Does the buzzer care about polarity like an LED?":

- "Yes, it has a + and – side" (correct for most piezo buzzers — they do have polarity)
- "No, it works either way" (partially true for some buzzers — encourage them to test it)
- "I'd test it by flipping it around and seeing if it still buzzes" (excellent experimental thinking — affirm this)

📷 **Capstone Photo Moment!** Build your buzzer + LED parallel circuit and take a photo. This photo should show both components clearly visible on the snap grid with the switch in the "on" position. Label it: "Parallel Buzzer + LED Alarm Circuit — Energy Conversion Demo."

✏️ Easter Egg Design — Engineering Notebook Entry

Before you build, every good engineer draws their plans. This isn't just a school requirement — NASA engineers, Apple hardware designers, robotics teams — they ALL draw circuit diagrams before they build anything physical. Your notebook is your engineering journal, and this entry is going to be one of the most important ones of the whole unit. Take it seriously and make it detailed.

🗣️ **Teacher Script — The Easter Egg Challenge:** "Alright, engineer. Here's your mission. I want you to create an Easter egg that: (1) lights up with at least TWO LEDs, (2) has a way to turn the lights on and off, and (3) here's the twist — one LED should be brighter than the other. Your job is to figure out HOW to make one brighter than the other using what you know about series and parallel. Hint: it involves resistance. You have 10 minutes to design your circuit on paper BEFORE you touch any components. Go."

✏️ **Drawing Prompt — Engineering Blueprint:** In your science notebook, create a formal circuit diagram for your Easter Egg Light project. Your diagram MUST include: (1) Battery symbol with + and – labeled, (2) Every component with its standard electrical symbol (the ones you drew earlier!), (3) Wire connections shown as straight lines with corners — no curved sloppy lines, (4) Labels on each component (R1, D1, D2, S1, etc.), (5) Arrows on the wires showing the direction of conventional current flow, (6) A short written explanation (2–3 sentences) next to the diagram explaining your strategy for the challenge: identify which LED will be brighter and why your chosen circuit layout (series/parallel/combination) achieves this effect. Title this page: "Easter Egg Light Circuit — My Engineering Design."

What Makes a Good Circuit Diagram

- Straight lines only for wires (use a ruler!)
- Standard symbols, not drawings of the physical parts
- Components evenly spaced for readability
- Clear labels with component codes
- Current direction arrows
- Title, date, and your name at the top

Common Mistakes to Avoid

- Drawing the physical snap piece instead of the symbol
- Forgetting to close the loop (open circuit — won't work!)
- Not labeling + and – on the battery
- Putting an LED in backwards in the diagram
- Forgetting the switch (it's easy to omit!)
- Making the diagram so small you can't read it

📷 **Capstone Photo Moment!** Take a photo of your engineering diagram BEFORE you build. Then take a photo of your FINAL built circuit. These two photos together tell a powerful story for your capstone: "I designed it, then I built it." Label them: "Easter Egg Circuit Diagram" and "Easter Egg Final Build."

🥚 Project 5 — Easter Egg Light Challenge

Now we get creative. The Easter Egg Light set gives us something we haven't had yet: a light-up display component that you can encase inside the Easter egg shell. This project is less about following instructions and more about making design decisions like an engineer. You have the knowledge now — series, parallel, switches, LEDs — so I'm going to give you a goal and you're going to figure out the circuit design yourself. This is the first time today where there's no step-by-step guide.

🗣️ **Teacher Script — The Challenge Briefing:** "Alright, engineer. Here's your mission. I want you to create an Easter egg that: (1) lights up with at least TWO LEDs, (2) has a way to turn the lights on and off, and (3) here's the twist — one LED should be brighter than the other. Your job is to figure out HOW to make one brighter than the other using what you know about series and parallel. I'll give you a hint: it involves resistance. You have 10 minutes to design your circuit on paper BEFORE you touch any components. Go."

1. Design Phase (10 min)

Draw your circuit diagram in your notebook before building. Label every component, mark + and – terminals, and write your strategy for making one LED brighter than the other. This is your engineering blueprint — no touching components until this is done.

2. Build Phase (8 min)

Snap together your designed circuit using Easter Egg Light set components plus anything from the other sets. No peeking at any guide — use YOUR diagram as the only reference.

3. Test & Iterate (5 min)

Does it work as planned? If not — troubleshoot! Trace your circuit with your finger, check polarity, verify all snaps are secure. Modify your design if needed and note what you changed.

4. Present & Explain (3 min)

Explain your final circuit to me out loud: trace the current path with your finger, name each component, describe each energy transformation, and explain WHY one LED is brighter than the other.

🔒 **Question for after the build to reaffirm knowledge**

💡 **Possible Student Approaches & Answers:**

For "How do you make one LED brighter than the other?":

- "Put one LED in series with a resistor and the other without" (correct — the one with the resistor gets less current and is dimmer)
- "Put one in parallel and one in series" (creative — this can work! Affirm the thinking and let them explore)

⚡ Going Deeper — Resistance, Brightness, and Ohm's Law Live

We talked about Ohm's Law in PhET. Now let's make it real. Ohm's Law says V = IR, which means Voltage = Current × Resistance. If you keep voltage the same (same battery) and increase resistance, current goes DOWN. Less current flowing through an LED means it's dimmer. This is the secret to making one LED brighter than another — and you can actually see it happening in front of you right now.

This is why adding a resistor in series with an LED makes it dimmer — you're increasing total resistance, decreasing current, and therefore decreasing the energy delivered to the LED per second. The LED is still getting the same voltage drop, but less current means less light output. Now go back to your Easter Egg circuit and verify this is actually what's happening in your build!

📢 **Teacher Script — Ohm's Law Challenge:** "Okay, here's a real calculation. Your AA batteries give us about 3 volts total. Your LED needs approximately 0.02 amps (20 milliamps) to light up properly. Using V = IR, calculate: what size resistor should we put in series to protect that LED? Work it out on paper. Show me your math." **[Answer: R = V/I = 3/0.02 = 150 ohms. A 100–200 ohm resistor is standard for LED protection. Celebrate when they get this!]**

💡 **Possible Student Answers:**

- "R = 150 ohms" (correct — celebrate this!)
- "R = 3 × 0.02 = 0.06 ohms" (common error — multiplied instead of divided; guide: rearrange V=IR to R=V/I)
- "I'm not sure how to rearrange the formula" (common — walk through: if V=IR, then R=V/I, cover the R in the triangle)
- "60 ohms" (calculation error — recheck the division: 3 ÷ 0.02 = 150, not 60)
- "Does it have to be exactly 150?" (great practical question — affirm: no, 100–200 ohm range is fine for most LEDs)

🔬 Activity 1 — Ohm's Law in Action: Watch Resistance Change Brightness

🔬 Activity 1 — Ohm's Law in Action: Watch Resistance Change Brightness

Using your Skill Builder set, build this circuit: Battery holder (B1) → Switch (S1) → Variable Resistor VR1 → LED (D1) → back to battery negative terminal. This is a simple series circuit — the only new component is the VR1.

1. Turn the VR1 dial all the way to minimum resistance. Flip the switch. The LED should glow at full brightness.
2. Slowly turn the VR1 dial to increase resistance. Observe what happens to the brightness.
3. Keep turning until the LED goes out completely.

What you're seeing is Ohm's Law live: as resistance increases, current decreases ($I = V/R$), and less current means less energy delivered to the LED per second — so it gets dimmer. When current drops below the LED's minimum threshold, it goes out entirely.

✏️ **Record in your notebook: At what point (roughly) did the LED start to noticeably dim? Did it fade gradually or cut out suddenly at the end? What does that tell you about the LED's voltage threshold?**

🎨 LED Colors & Voltage — Why Does Color Matter?

Earlier in this lesson, when your LED first lit up, we mentioned that the color of light depends on how big the energy drop is inside the semiconductor material. Now that you understand Ohm's Law and voltage, we can go deeper. It turns out that different colored LEDs don't just look different — they actually need different amounts of voltage to light up at all. This is Ohm's Law and photon physics working together in real time.

> 📢 **Teacher Script:**
> 'Remember back on page 7 when we talked about photons — how electrons drop to a lower energy level and release that energy as light? We said the color depends on how big the energy drop is. Well, here's the connection to what we just learned: bigger energy drop = higher energy photon = blue or violet light. Smaller energy drop = lower energy photon = red light. And here's the kicker — a bigger energy drop requires MORE voltage to push the electron up to that higher level in the first place. So blue LEDs need more voltage than red ones. You can actually test this with your kit right now.'

🔴 Red LED - Smallest energy drop. Lights up easiest. Works well even with a weak or partially drained battery.

🟠 Orange/Yellow LED - Slightly larger energy drop. Needs a bit more push than red.

🟢 Green LED - Middle range. Standard green LEDs need around 2.2V to light up reliably.

🔵 Blue LED - Large energy drop. Needs significantly more voltage — nearly double a red LED!

🏆 Blue LED Fun Fact - The blue LED was so hard to make that its invention won the Nobel Prize in Physics in 2014. Scientists spent decades trying to figure out the right material.

🎨 Activity 2 — Color vs. Voltage: The Threshold Test

Now that you've seen the voltage chart, it's time to test it. We're going to connect three different colored LEDs in PARALLEL using a breadboard, jumper wires, and alligator clips, then use the 10K variable resistor to slowly reduce current until they cut out one by one — in color order. Watch the voltage chart come to life in real time.

🔌 **Setup Guide — Connecting Loose LEDs via Breadboard, Jumper Wires & Alligator Clips**

You'll need: 3 loose LEDs (red, green, yellow), 6 short male-to-male jumper wires, 2 alligator clip leads, your 400-hole half-size breadboard, and your Snap Circuits board with Battery holder (B1), Switch (S1), and 10K Variable Resistor (VR1) already built from Activity 1.

📌 Quick Breadboard Reminder: On your breadboard, the NUMBERS run horizontally left to right — these are your ROW numbers. The LETTERS run vertically top to bottom — these are your COLUMN letters. Same numbered row = internally connected (e.g., 10a, 10b, 10c, 10d, 10e are all connected). Different numbered rows in the same column = NOT connected (e.g., 10a and 11a are NOT connected).

📌 Reference Diagram — use this as your visual guide for the setup steps below.

Step 1: Leave the VR1 circuit from Activity 1 intact. Remove the D1 snap LED from the end of the circuit.

Step 2 — Insert and connect the LEDs into the board:

🔌 Activity 2 — LED Wiring & Power Connection

Step 2 continued — Insert each LED into the breadboard:

🔴 RED LED

Long leg (+) → 10a

Short leg (–) → 11a

Positive jumper → 10b → red rail

Negative jumper → 11b → blue rail

🟡 YELLOW LED

Long leg (+) → 13a

Short leg (–) → 14a

Positive jumper → 13b → red rail

Negative jumper → 14b → blue rail

🟢 GREEN LED

Long leg (+) → 16a

Short leg (–) → 17a

Positive jumper → 16b → red rail

Negative jumper → 17b → blue rail

Step 3 — Connect each LED to the power rails with jumper wires (already configured in columns above). Each jumper plugs into column b of the same row as the LED leg — column b is in the same row as column a, so they are internally connected.

Step 4 — Connect Snap Circuits to the power rails:

- Alligator clip → anywhere on the red power rail → Snap Circuits positive output snap connector
- Alligator clip → anywhere on the blue ground rail → Snap Circuits negative/ground snap connector

📌 It doesn't matter which hole on the rail you use — the entire red rail is one continuous connection, and the entire blue rail is one continuous connection. Any hole works!

Step 5 — Double-check before powering on:

- Each LED long leg shares a row with its positive jumper ✅
- Each LED short leg shares a row with its negative jumper ✅
- No two LED legs share the same row as each other ✅
- Long leg = positive, short leg = negative ✅
- Red power rail → Snap Circuits (+) ✅
- Blue ground rail → Snap Circuits (–) ✅

⚠️ Important: Make sure all LED legs and jumper wires stay in column a or b — do NOT straddle the center gap of the breadboard.

🔬 Activity 2 — The Threshold Test

🔬 The Threshold Test — Watch the Color Order

Now for the moment of truth.

1. Turn VR1 to minimum resistance. Flip the switch. All three LEDs should glow.
2. Write your prediction in your notebook: which color will cut out FIRST? Which will be LAST?
3. Slowly turn the VR1 dial to increase resistance. Watch carefully.
4. Note which color cuts out first, second, and last. Record the order in your notebook.

Expected result: green cuts out first → yellow next → red last. This matches the voltage chart exactly — green needs the most voltage, red needs the least.

✏️ **Record in your notebook:** Create a table — Color | Cut Out Order (1st/2nd/3rd) | Matches Chart Prediction? (Yes/No). Then write 2–3 sentences explaining WHY the colors cut out in that order using the words: voltage threshold, energy drop, photon, and resistance.

📢 Teacher Script — Connecting It All

'So think about what just happened. You used a variable resistor to control how much current flows. Less current means less voltage drop across the LED. And when that voltage drop falls below the LED's threshold — no more photons. The color of the light isn't just aesthetic — it's physics. You might be wondering why we're not using a blue LED today — blue LEDs actually need MORE voltage than our batteries can reliably deliver when shared. That's a real engineering constraint you just ran into. With only 2 AA batteries (3V total), a blue LED's forward voltage of about 3.0–3.5V is right at or above the battery limit, so it would barely glow or not light at all. Yellow LEDs need only about 2.0–2.2V, which works much better in this setup.'

🔒 **Teacher Reference Only — Do Not Read Aloud** Possible Student Reactions:

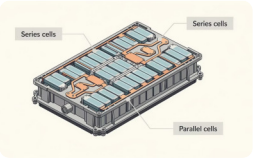
- 'Wait, the blue LED really needs that much more voltage?' (yes — affirm: blue LEDs need about 3.0–3.5V, which is too close to the battery limit when current is shared across multiple LEDs)
- 'Why does it cut out instead of just getting dimmer?' (great question — there's a minimum threshold voltage; below it, electrons can't make the full energy jump at all)
- 'So all LEDs have a threshold?' (correct — every LED color has a forward voltage minimum)
- 'What about white LEDs?' (optional bonus fact — white LEDs are often made from a blue LED with a phosphor coating that shifts some of the light, but that's just extra trivia for later!)

🌍 Real-World Connections — Where Do You See This Every Day?

Everything we've built today isn't just a science kit activity — it's the foundation for almost every piece of technology that exists. Let's zoom out for a second and connect snap circuits to the real world. Understanding these connections is what makes science meaningful, not just "stuff you do in school." Every device you interact with daily contains these exact principles.

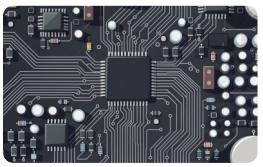
Your Home's Electrical System

Your entire house is wired in parallel. Every outlet, every light switch, every appliance is its own branch. That's why you can turn off one lamp without turning off your TV, and why a blown fuse in the kitchen doesn't kill the living room lights. Circuit breakers are just big switches that open the circuit automatically when too much current flows — exactly like your slide switch, just much bigger and automatic.



Electric Vehicles and Batteries

The Tesla Model 3 battery pack has over 2,000 individual battery cells. They're wired in a COMBINATION of series and parallel: cells in series increase the total voltage, while cells in parallel increase the total current capacity. The engineers who designed that battery pack were using the exact same principles you used today — just with thousands of components instead of five.



Inside a Computer or Smartphone

Your phone's circuit board has billions of transistors — essentially microscopic switches, just like your S1 switch but a million times smaller. The CPU controls the flow of current through countless tiny paths, switching billions of times per second. Every calculation your computer makes is fundamentally electricity flowing through switches arranged in logical patterns — the same concept as your simple switch circuit, just almost incomprehensibly miniaturized.

🎯 Synthesis Challenge — Design Your Own Circuit

(Only if time allows)

This is the culminating challenge of the Snap Circuits lesson. You've built simple circuits, series, parallel, motor, buzzer, and the Easter Egg. Now I want to see you combine everything into ONE design of your own choosing. This challenge has constraints — because in real engineering, you always have constraints. The best engineers create brilliant solutions WITHIN limits, not without them. Here are your constraints for this final challenge.

🔧 Constraint 1 — Components

Your circuit must use at least 5 different types of components (not 5 of the same thing — 5 different types). You may draw from ANY of the four sets: Snaptricity, Skill Builder, Snap Circuits Junior 100, Easter Egg Light.

⚡ Constraint 2 — Circuit Type

Your circuit must include at least one section that is in PARALLEL. You may also include series sections — in fact, a combination is encouraged because it's more interesting and more challenging.

💡 Constraint 3 — Energy Transformations

Your circuit must demonstrate at least TWO different energy transformations from electrical energy (e.g., electrical → light AND electrical → sound, or electrical → light AND electrical → mechanical motion).

📋 Constraint 4 — Documentation

You must produce: (a) a complete circuit diagram drawn BEFORE building, (b) a component list with function notes, and (c) a written explanation of why you made your key design choices (minimum 3 sentences).

🗣️ **Teacher Script:** "There's no right answer here. I'm not looking for a specific circuit — I'm looking for evidence that you can think like an engineer. Your circuit doesn't have to be complex to be excellent. A simple circuit that's perfectly understood and clearly documented beats a complex circuit you can't explain every single time. Go design something you're proud of."

📷 **Capstone Photo Moment!** This is your biggest capstone photo. Take multiple shots: (1) your planning diagram, (2) your circuit mid-build, (3) your final circuit fully lit/operating. These three images together are perfect for a capstone "design process" slide. Label them: "Synthesis Challenge — Plan," "In Progress," and "Final Design."

✏️ Lesson Wrap-Up — The Big Ideas Note

Before we transition, we're going to lock in the big ideas from this entire lesson. This is your most important notebook entry of the day — a synthesis of everything you've learned, in your own words. This note will be the first thing you look at before any circuits test or project presentation, so make it good. Write like you're explaining it to your future self who forgot everything.

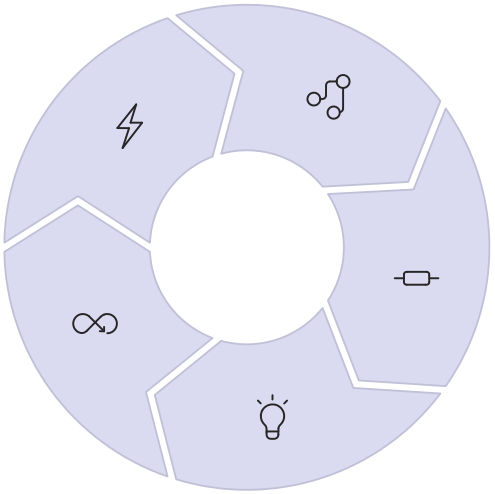
✏️ **Comprehensive Note-Writing Prompt — "The Big 5 of Circuits":** Create a structured notebook section titled "Snap Circuits — Key Takeaways." Include these five sections, each with 3–5 sentences IN YOUR OWN WORDS plus a small sketch: (1) What a circuit needs to function — the essential conditions for current flow. (2) Series vs. Parallel — the key difference and one real-world example of each. (3) Ohm's Law — what V = IR actually means and how you saw it in action today. (4) Energy transformations — what electrical energy turned into in each project you built. (5) Troubleshooting — your personal top 3 "check these first" steps. At the bottom, write one question you still have about circuits — something you're still curious about or not 100% sure of.

Voltage Drives

Chemical energy in batteries creates voltage — the "electrical pressure" that pushes current

Current Flows

Electrons move through the complete loop from negative to positive terminal



Resistance Limits

Components convert electrical energy and slow current — the balance creates useful output

Energy Transforms

Electrical energy becomes light, heat, sound, or motion in each component

Loop Returns

Current completes the journey back to the battery, ready to be pushed around again

📐 ISTE Standards Coverage — This Lesson

This lesson isn't just meeting state science standards — it's also addressing multiple ISTE (International Society for Technology in Education) standards for students. These are the technology and digital-age learning standards that describe what it means to be a capable, thoughtful technology user and creator. Here's exactly which standards this lesson covers and how:

ISTE 1.4 — Innovative Designer

Students use a variety of technologies within a design process to identify and solve problems by creating new, useful, or imaginative solutions. **Covered by:** The Easter Egg Light Challenge and Synthesis Challenge require the student to go through a full engineering design process — defining constraints, designing solutions, building prototypes, testing, and iterating. The circuit diagram requirement mirrors professional design documentation practices.

ISTE 1.5 — Computational Thinker

Students develop and employ strategies for understanding and solving problems in ways that leverage the power of technological methods to develop and test solutions. **Covered by:** The systematic troubleshooting protocol (Section 17) directly trains computational thinking — breaking a complex problem into steps, isolating variables, and testing hypotheses systematically. Series vs. parallel comparisons use logical analysis to understand system behavior.

ISTE 1.6 — Creative Communicator

Students communicate clearly and express themselves creatively for a variety of purposes using the platforms, tools, styles, formats, and digital media appropriate to their goals. **Covered by:** Circuit diagrams, capstone photos, engineering notebook entries, and the final presentation-ready documentation all develop the student's ability to communicate technical ideas through multiple formats including visual, written, and photographic documentation.

ISTE 1.3 — Knowledge Constructor

Students critically curate a variety of resources using digital tools to construct knowledge, produce creative artifacts, and make meaningful learning experiences. **Covered by:** The explicit connection between PhET simulation (digital tool) and physical Snap Circuits (hands-on tool) requires the student to construct knowledge by synthesizing across two very different learning modalities. The student builds a personal reference notebook that is a curated knowledge artifact.

ISTE 1.7 — Global Collaborator

Students use digital tools to broaden their perspectives and enrich their learning by collaborating with others and working effectively in teams. **Covered partially by:** The real-world connections section (Section 18) and the transition to micro:bit programming both position the student within a global community of electronics engineers and makers who use these exact skills to solve real-world problems. The capstone documentation prepares work for sharing and presenting to broader audiences.

🚀 Transition Script — From Snap Circuits to Micro:bit

This is it — the bridge to your next adventure. Everything you've built today was the foundation. Snap Circuits showed you how physical components work in the real world. Now we're going to take that knowledge and add a completely new superpower: **programmable control.** Welcome to the world of microcontrollers.

🗣️ Teacher Script — Read This Out Loud to Close Snap Circuits and Open Micro:bit:

"Okay. Look at what you built today. You made light. You made motion. You made sound. You designed your own circuit from scratch. That's legitimately impressive — most adults can't do what you just did. But here's my question for you: everything you built today, how did you turn it on? A switch, right? A physical switch you had to flip with your finger.

What if I told you that the SWITCH could be code? What if instead of flipping a physical piece of plastic, you could write instructions — like a program — that decides WHEN the circuit turns on, for HOW LONG, in WHAT PATTERN, and in RESPONSE TO WHAT? A temperature reading. A button press. A hand gesture. A light sensor.

That's what a microcontroller does. It's basically a tiny computer on a chip that reads inputs from the world — sensors, buttons, light detectors — and then makes decisions about what outputs to activate. Sound familiar? Because it's EXACTLY what you were doing manually with your switch and your hand. The micro:bit doesn't change the circuit fundamentals you learned today. It replaces your hand.

And here's the coolest part: the LED, the motor, the buzzer — all those components you just learned? You can connect them to a micro:bit. The skills aren't separate — they BUILD on each other. Everything you learned about series circuits, voltage, current, and components is still 100% in play. We're just adding the ability to make the circuit think.

💡 Possible Student Answers — 'How did you turn everything on today?'

- "With the switch / by flipping the switch" (correct — this is the target answer the script is building toward)
- "By putting the batteries in" (also valid — guide: yes, but how did you control WHEN it turned on?)
- "By completing the circuit" (excellent conceptual answer — affirm and connect to the microcontroller idea)
- "By pressing a button" (correct for some builds — connect: a microcontroller can respond to button presses too, but also to sensors, timers, and code)

🎉 Capstone Recap — Your Photo Story

Before we dive into micro:bit, let's take 3 minutes to review the capstone photos you've taken today. This is your visual record of an entire circuits learning journey in one lesson. When you present your capstone, each of these photos tells a specific chapter of your engineering story. Here's the complete list of capstone moments from today — make sure you have all of them.

📷 Photo 1

Component Reference Sheet — "What I knew at the start"

📷 Photo 2

Project 1 LED glowing — "My first real circuit"

📷 Photo 3

Series vs. Parallel side-by-side — "Seeing the difference"

📷 Photo 4

Buzzer + LED Parallel Alarm — "Electricity makes sound AND light"

📷 Photo 5

Easter Egg Diagram + Final Build — "I designed this myself"

📷 Photo 6

Synthesis Challenge (Plan + In Progress + Final) — "Full engineering process"

Together, these photos document the full arc: from identifying components to designing original circuits. That's a genuinely impressive learning journey to showcase. Make sure each photo is labeled with what it shows, the date, and your name before you save it to your capstone folder. Your future self — and your audience — will thank you for the clear labels.

> 📢 **Final Teacher Script:** "Look at everything you did today. This morning, electricity was something that came out of a wall socket and made your phone charge. Now? Now you KNOW how it works. You've built circuits that prove the fundamental laws of electricity aren't just textbook concepts — they're real, they're predictable, and when you understand them, you can make electricity do exactly what you want it to do. That's engineering. That's science. And that's exactly the foundation you'll need when we start programming the micro:bit. Let's go."